

AllegSkill

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AllegSkill

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Technical Details: Commander's ranking · **Player's ranking** · Stack rating · AllegBalance

AllegSkill is a system for rating the skill of Allegiance players based on their overall performance in-game. AllegSkill is based on the Trueskill (<http://research.microsoft.com/en-us/projects/trueskill/default.aspx>) system developed by Microsoft Research (<http://research.microsoft.com/default.aspx>) (who also developed Allegiance) with some notable additions. The term 'AllegSkill' is intended to refer to the entire system, which includes additional statistics, and Microsoft Research should not be held responsible for differences when and where the occur. This wiki will deal principally with the technical aspects of the system. For a layman's explanation of the basics, please see the Trueskill (<http://research.microsoft.com/en-us/projects/trueskill/details.aspx>) site.

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How it works

You have two numbers keeping track of your rank: Mu, μ , and Sigma, σ . μ is your rating (your average skill you've exhibited in all the games you play), and σ is the *uncertainty* about your rating. After you play a game your μ goes up if you win, down if you lose. Your σ always goes down.

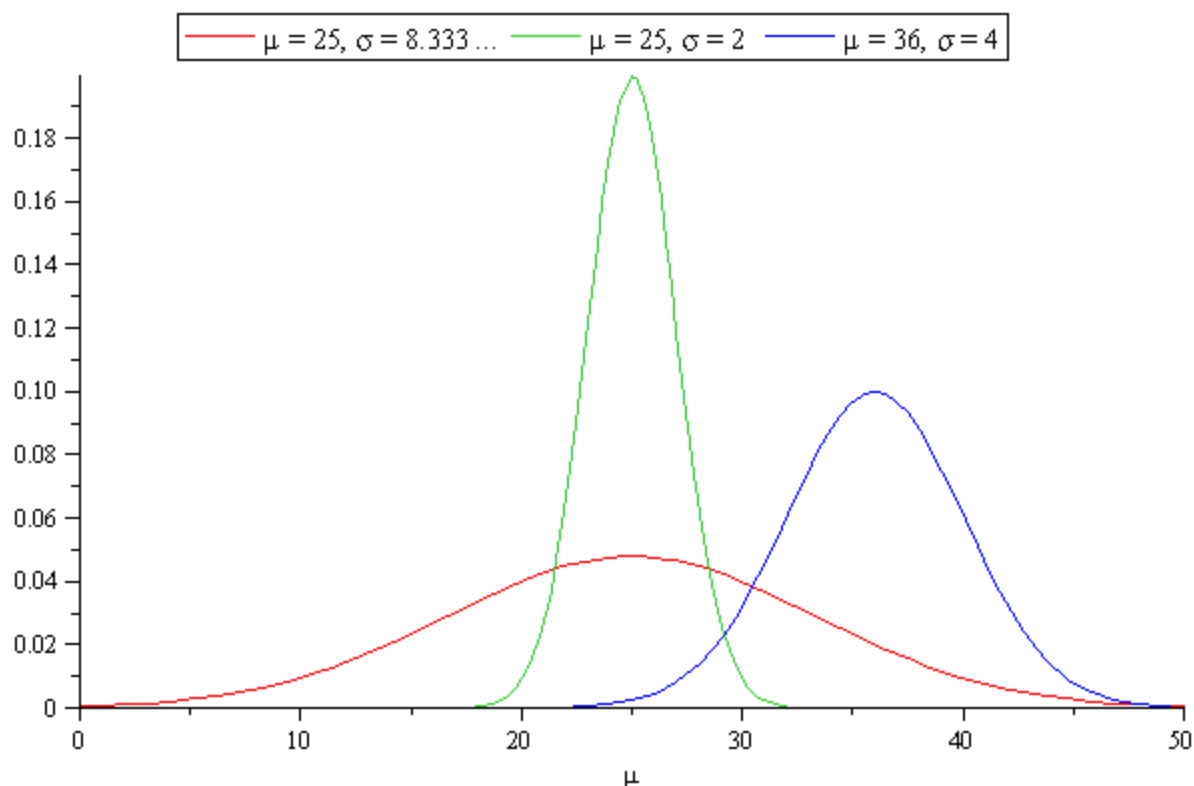
The amount your μ and σ go up/down is modified by σ (the more certain the system is about your rank, the less each game will affect it) and by how 'surprising' the game outcome was (a newbie beating a veteran is quite surprising and will have a greater effect on the ranks than if the vet beat the newb).

AllegSkill realises that whether a team wins or loses is highly dependant on the skill of both the commanders and their team, and the algorithms used represent this. Consequently there are separate skill ratings for commanders and pilots.

Mu & Sigma

As mentioned previously, mu and sigma represent the average skill of a player and the uncertainty around that skill respectively. Uncertainty is a fundamental part of the AllegSkill system, and deserve greater explanation. We have chosen to plot a graph of three players with different skill ratings:

1. A **newbie** player.



- Average rating (25) and high uncertainty (8.333).
- 2. A **skilled player that has played only a little**
 - Higher rating (36) and moderate uncertainty (4)
- 3. An **average player that has played lots**
 - Average rating (25) and low uncertainty (2)

The horizontal axis represents their rating, and the vertical axis is the probability. A simple way of interpreting these graphs is to think that the players "true rank" lies *somewhere* between the two end points of the curve and that the higher the curve is, the more likely the true rating is at that point.

So, to interpret our three players curves:

1. The red line touches the mu-axis at 0 and 50. This means that the AllegSkill system believes that the player could have any skill between these two values.
 - All new players start out with this skill rating.
2. The blue line touches 22 and 50, so this player is either just below average, or the best expert in the game.
3. The green curve hits the mu-axis between 18 and 32, but is quite likely to be 25.

Stable uncertainty

Note that players stabilise with a σ of approximately 1. This would result in a skill distribution even 'tighter' than that represented by the **green** player.

By itself, σ would eventually decrease to zero. However γ is the dynamics variable which prevents sigma from ever reaching zero, which in turn determines how quickly mu can in/decrease once sigma has stabilised. If we discover that sigma-stabilised ratings are moving too slowly to reflect genuine changes in skill, we will increase gamma. It is because of gammas that some player's sigma are less than 1.

Your rank

The rank that is displayed in-game is known as your **Conservative rank**. Basically it is where your curve touches the axis on the left - the system is 99% sure that your "true rank" isn't any *less* than your conservative rank.

The formula is

$$\text{Conservative rank} = \mu - 3 \times \sigma$$

This means that newbies, who start with $(\mu, \sigma) = (25, 8.33)$ will have a conservative rank of zero. As they play more games the uncertainty about their rank goes down, and so their conservative rank more closely resembles their rating - which will hopefully be close to resembling their actual skill by the time they've played that many games!

Leaderboard/Ingame ranks

The ranks shown on leaderboard/in game are multiplied by 0.6, so we have ranks up to 30, instead of ranks up to 50. This is a temporary measure to ease the process of migrating from Helo's 0-30 system to AllegSkill's 0-50 system. Once the various non-ASGS components of Allegiance (game servers etc) have been updated to fully support AllegSkill (with additional features such as a new autobalance system), we will switch over to 0-50 ranks across the board. It is also worth noting that currently the 'rank' column on the leaderboard is also multiplied by 0.6, so should accurately reflect the rank displayed in-game.

The formula is

$$\text{Leaderboard rank} = (\mu - 3 \times \sigma) \times 0.6$$

Related Articles

- Why is HELO broken and can it be repaired?

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